

Current conclusion display preservation

Cyclic connection and flatness

$$\begin{aligned}\Omega_u &= -C_\chi(t)/u^2 + B_\chi/u, & \Pi_u &= \Pi\Omega_u, \\ \Omega_t &= -A(t)/u, & \Pi_t &= \Pi\Omega_t, \\ \partial_u\Omega_t - \partial_t\Omega_u + [\Omega_u, \Omega_t] &= 0.\end{aligned}$$

Residual collision annihilator and exact modules

$$\begin{aligned}\psi_{h,k}(X) &= \prod_{\mu} (X - \mu)^{\max_{\alpha}(\sum_i \rho_i) = \mu \min(p, 1 + \sum_i (m_i - 1))}, \\ \ker \bar{\eta}_k &= (\psi_{h,k})/(\bar{\chi}_{h,k}), \quad \text{im } \bar{\eta}_k \simeq k_0[X]/(\psi_{h,k}).\end{aligned}$$

Translated metric and all three pure derivatives

$$\begin{aligned}\psi_a - \psi &= 2p \operatorname{Re}(h_a(t)/u), \\ \partial_u(\psi_a - \psi) &= -ph_a(t)/u^2, \\ \partial_u^2(\psi_a - \psi) &= 2ph_a(t)/u^3, \\ \partial_u\partial_t(\psi_a - \psi) &= pa/u^2.\end{aligned}$$

Existing parent RMT16 overlay (unchanged)

$$\begin{aligned}\mathcal{H}_a(B) &= 2 \operatorname{arcosh}(e^{B/(2a)}), & \mathcal{H}'_a(B) &= \frac{1}{a\sqrt{1 - e^{-B/a}}}, \\ \mathcal{H}_a^{-1}(z) &= 2a \log \cosh(z/2) \quad (z > 0).\end{aligned}\tag{RMT16}$$